CSC2106: IoT Protocols and Networks
Project Proposal Report

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Project Title
LoRa Mesh network for Deep Tunnel Infrastructure Connectivity in Singapore

1. What specific issue or problem has been identified that requires a solution?
Singapore is currently undertaking large-scale deep-tunnel infrastructure projects, such as the Cross Island Line (CRL), with tunnels reaching depths of up to 70 meters, equivalent to a 25-storey HDB block. (Ministry Of Transport, 2019). LTA material repeats this depth framing.

A research study done by IEEE showed that ensuring reliable mobile connectivity in these deep tunnels is challenging due to their closed environments and reinforced concrete structures, which significantly attenuate radio signals through reflection, absorption, and lossy wall effects (Abdus S et al., 2023). Industry analyses of modern building materials further confirmed that rebar-reinforced concrete acts as a partial Faraday cage. Steel reinforcement grids reflect and absorb electromagnetic waves, causing signal losses of up to 30dB or more, equivalent to a 99.9% reduction in power for standard walls, with thicker tunnel walls causing significantly higher attenuation. (Resolution Wireless, 2025)

Current implementations rely on 4G and 5G signals distributed through repeaters, distributed antenna systems (DAS), and leaky feeder cables. However, technical comparisons indicate that leaky feeder systems suffer significant longitudinal signal loss, particularly at frequencies above 1 GHz, which are critical for modern 4G and 5G communications. Due to their linear, single polarized nature, typical leaky-feeder deployments provide limited spatial/polarisation diversity compared with antenna-based systems, and practical MIMO benefits can be constrained by tunnel geometry and feeder configurations.

As a result, achievable bandwidth is reduced and latency increased compared to antenna-based solutions. Furthermore, the reliance on multiple in-line amplifiers introduces faults that are difficult to detect, increasing system complexity and operational burden. (Poynting Antennas, 2023)

Additionally, measurement-based studies of 4G and 5G deployments show that while 5G particularly in mid and high-frequency bands can deliver very high throughput, its performance is significantly more sensitive to obstruction, distance, and penetration loss than lower-frequency systems. Millimeter-wave 5G, in particular, suffers rapid degradation under blockage such as building materials and human body interference, necessitating dense small-cell deployments to maintain performance. These propagation characteristics limit the reliability of high-frequency 5G in complex or obstructed environments compared to lower-band deployments. (Muhammad Iqbal R et al., 2023)

Another critical safety challenge lies in maintaining reliable wireless connectivity for environmental monitoring sensors, such as toxic gas and temperature detectors, without relying on extensive wired or leaky-feeder cabling infrastructure. In tunnel environments, such cabling systems are costly to install and difficult to modify once deployed, particularly as they are often located in hard-to-access areas. Moreover, exposure to high humidity, extreme temperatures, mechanical stress, and corrosion makes these systems vulnerable to degradation and faults that are difficult to detect and repair, increasing operational risk for safety-critical monitoring applications. (Peter B, 2024)

Radio propagation studies in curved tunnels also shows that bends introduce additional attenuation beyond straight-tunnel propagation, with electromagnetic energy tending to concentrate along the concave wall, a phenomenon often described using whispering-gallery-mode analogies. Non-line-of-sight propagation around tunnel bends experiences significant extra loss due to the rapid decay of higher-order modes, with the severity of attenuation increasing at higher frequencies. As frequency increases into the GHz range, signals become increasingly sensitive to curvature and obstruction, meaning that without intermediate repeaters, distributed antennas, or multi-hop mesh topologies, tunnel bends can create localized coverage gaps, leaving workers and environmental sensors situated beyond the bend in radio dead zones with no reliable connectivity. (Bo A et al., 2015)

Collectively, these findings indicate that conventional high-frequency cellular and cable-based systems are fundamentally mismatched to deep-tunnel environments, motivating the exploration of low-frequency, multi-hop wireless alternatives capable of adapting to tunnel geometry and structural attenuation.

2. Why is your topic important or relevant to industry, academia, or society?

This topic addresses critical infrastructure safety and operational efficiency challenges in Singapore's ongoing and future underground construction projects. Large-scale deep-tunnel systems, such as the Deep Tunnel Sewerage System (DTSS), represent long-term national infrastructure requiring high levels of reliability and operational assurance. DTSS Phase 2 involves the construction of a 98 km-long network of deep tunnels and link sewers serving the western half of Singapore, underscoring the scale and complexity of underground works and the need for robust monitoring of tunnelling activities and environmental conditions.
(PUB, 2025; Justin O, 2019)

Industrial relevance: The construction and infrastructure sector increasingly requires cost-effective alternatives to traditional wired communication systems. Studies of underground communication infrastructure indicate that leaky feeder and other wired systems are dominated by high cable, installation, and maintenance costs and are difficult to reconfigure once deployed. Industry analyses report that installed wired instrumentation can cost approximately USD $10–20 per foot, with even short conduit runs of 50–75 feet costing up to USD $1,500, while wireless monitoring systems can reduce on-site installation labor by 50–75% by eliminating trenching and conduit works and enabling flexible, scalable expansion as construction requirements evolve. (Dana T, 1980; BBP, n.d.)

Academic relevance: Prior research in underground mining and tunnel communications has identified multi-hop LoRa mesh networks as a promising solution for extending coverage in subterranean environments where conventional LoRaWAN deployments are impractical. However, existing studies are largely limited to mining or linear tunnel scenarios and rely predominantly on simulation or simplified flooding-based approaches, with limited comparative analysis of dense LoRa mesh performance in complex civil infrastructure settings. This project addresses this gap by investigating LoRa mesh performance in vertically constrained and structurally obstructed underground environments, contributing new insights relevant to urban deep-tunnel infrastructure contexts. (Nalith U et al., 2025)

Societal Impact: Under Singapore's MOM Workplace Safety and Health (WSH) Regulations, real-time gas monitoring is mandatory for confined and underground work environments where toxic or flammable gases may be present. In deep-tunnel infrastructure projects such as MRT and DTSS, deploying continuous monitoring is challenging due to vertical depth and structural obstructions. (Minerva, n.d.) A LoRa mesh–based wireless sensor network enables MOM-compliant, real-time gas detection without reliance on fixed cabling, maintaining connectivity around tunnel bends and shafts. This improves emergency response coordination and significantly enhances worker safety in Singapore's underground construction environments.

3. What is the potential impact of addressing this challenge?

The successful implementation of a LoRa-based mesh or multi-hop wireless network within Singapore's deep-tunnel infrastructure represents a significant shift away from rigid, cable-dependent communication systems towards a more resilient, flexible, and cost-efficient connectivity model. By directly addressing the propagation, installation, and maintenance challenges inherent in subterranean environments, this approach enables a communication architecture that is better suited to the complex geometry and harsh conditions of deep underground infrastructure.

Economic Impact and Infrastructure Cost Efficiency: One of the most significant impacts of adopting a LoRa-based wireless sensor network is the reduction in installation and lifecycle costs associated with underground monitoring systems. Conventional wired solutions rely heavily on extensive cabling, conduits, and repeaters, which incur high material, labour, and

maintenance costs that scale linearly with tunnel length (MDPI, 2024). In large-scale tunnel projects spanning several kilometres, such costs quickly become prohibitive.

By eliminating the need for leaky feeders and extensive hardwired connections, a wireless LoRa-based architecture significantly reduces both upfront installation expenses and long-term maintenance requirements. Owing to its long-range, low-frequency propagation characteristics, LoRa outperforms WPAN and WLAN technologies such as ZigBee and Wi-Fi in underground environments, requiring fewer gateways or relay nodes to achieve comparable coverage (IEEE, 2024; MDPI, 2024). This reduction in infrastructure complexity directly translates into improved cost efficiency and lower operational expenditure over the lifespan of the tunnel system.

Operational Agility and Scalability: Addressing underground connectivity challenges using a LoRa-based mesh architecture introduces operational flexibility that is difficult to achieve with traditional linear systems such as leaky feeders. LoRa-based wireless sensor networks are inherently scalable, allowing additional sensor nodes or relay devices to be deployed incrementally as tunnel excavation progresses, without requiring major modifications to existing infrastructure (MDPI, 2024).

This flexibility enables a “plug-and-play” deployment model, where sensors can be repositioned or added to accommodate evolving construction phases and monitoring requirements. Although LoRa trades off data rate to achieve long-range, low-power communication, this characteristic is well suited for environmental monitoring applications, where periodic measurements must balance responsiveness with battery longevity (ScienceDirect, 2023). As a result, the system supports continuous monitoring while minimising maintenance effort and power consumption.

Mission-Critical Safety and Disaster Resilience: The most critical impact of this project lies in its contribution to workplace safety and regulatory compliance. By extending reliable connectivity into areas previously affected by signal dead zones caused by tunnel curvature, reinforced concrete, or depth, the proposed network enables real-time monitoring of toxic gases and environmental conditions in accordance with Singapore’s Workplace Safety and Health (WSH) regulations.

Unlike wired systems, which are vulnerable to failure due to humidity, corrosion, mechanical damage, or explosions, a wireless mesh-based architecture offers inherent redundancy and self-healing capabilities. Through multi-hop communication, sensor data can be dynamically rerouted if individual nodes are damaged or obstructed, maintaining connectivity between underground sensors and surface control rooms (IEEE, 2024; MDPI, 2024). This resilience enhances early warning capabilities for hazardous conditions and supports predictive maintenance strategies, ultimately improving worker safety and reducing operational risk in deep-tunnel environments.

4. What specific gap or challenge in the field does this work address?

While prior research has demonstrated the feasibility of LoRa-based communication in underground environments such as mines and linear utility tunnels, significant gaps remain when these approaches are applied to Singapore's deep-tunnel civil infrastructure context.

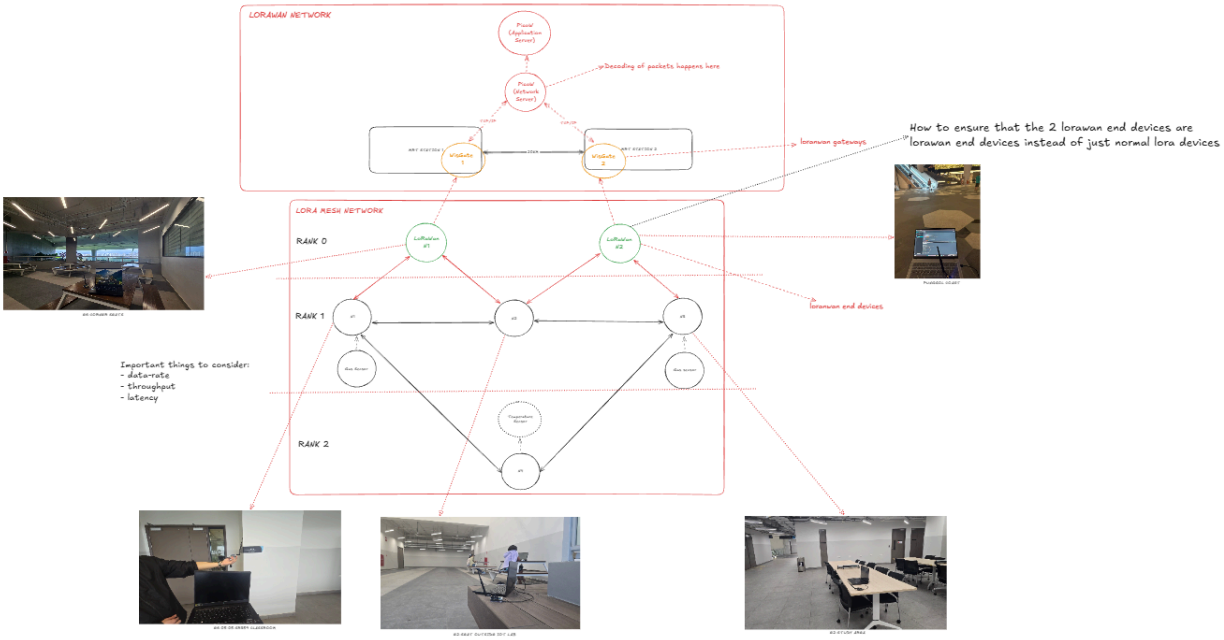
First, existing studies on underground LoRa deployments are predominantly conducted in mining corridors or straight tunnel geometries, where line-of-sight or near-line-of-sight propagation is achievable between nodes. Singapore's deep-tunnel infrastructure, such as the Cross Island Line and DTSS, involves curved tunnel sections, vertical shafts, and structurally complex environments where tunnel bends introduce additional attenuation beyond straight-tunnel propagation (CHOOVIO, 2025). These geometrical characteristics create radio dead zones that flooding-based or single-hop LoRaWAN architectures cannot reliably overcome, yet there is limited empirical study of LoRa mesh performance specifically in these vertically constrained, curved civil tunnel environments.

Second, existing underground wireless sensor network deployments rely on homogeneous sensor payloads, a single sensor type per deployment. (ByungWan J. et al, 2018, Mohd A. et al, 2017) In safety-critical tunnel environments requiring simultaneous monitoring of toxic gases, temperature, and potentially structural vibration, the network must accommodate heterogeneous data sources with varying packet structures and urgency levels. (Syed Jalal A. et al., 2022, Gaohua Z, 2025) Current LoRaWAN implementations are largely payload-aware at the gateway level, creating a tight coupling between sensor type and network configuration. This makes incremental sensor deployment critical in an active construction environment where monitoring requirements evolve as excavation progresses, operationally complex and costly.

Lastly, while self-healing mesh topologies have been proposed for underground resilience, there is limited work evaluating how payload-agnostic forwarding at relay nodes affects end-to-end latency and channel efficiency in a multi-hop LoRa mesh under real tunnel attenuation conditions (Kennedy, 2006). For WSH-compliant gas monitoring, alert delivery latency is safety-critical (WSHCouncil, n.d.) yet no study has directly quantified the latency benefit of decoupling the forwarding layer from payload inspection in a resource-constrained LoRa mesh node operating underground.

This work addresses these gaps by designing and evaluating a payload-agnostic LoRa mesh network architecture tailored to Singapore's deep-tunnel infrastructure, measuring the impact on per-hop forwarding latency, channel efficiency, and deployment scalability across heterogeneous sensor types in a structurally obstructed environment.

5. What are the main goals of this project or research?



Multi-tier LoRa mesh for MRT tunnels

The first goal of this project is to design and implement a two-level LoRa mesh architecture tailored for Singapore's deep MRT tunnel environment, in which Level 0 sensor nodes form a lateral mesh along the tunnel and Level 1 LoRaWAN bridge nodes relay this traffic to gateways located at adjacent MRT stations. By emulating the geometry and reinforced-concrete structure of Cross Island Line–style tunnels, including bends and vertical shafts, this architecture directly addresses the deep-tunnel propagation and coverage gaps highlighted in Question 1, while supporting the flexible, cable-light deployment model and operational cost savings discussed in Questions 2 and 3.

Load-aware routing and stress testing

The second goal is to develop a routing protocol that supports reliable multi-hop forwarding under realistic MRT tunnel traffic conditions, and to subject it to stress tests focusing on throughput and latency. Each node maintains path metrics such as hop count, link quality and queue occupancy so that traffic can be steered away from congested relays, distributing packets across multiple paths towards the WisGate gateways at each station and preventing single-node overloads. By injecting bursty gas and temperature telemetry that emulate incident scenarios in a curved, obstructed pathway similar to a tunnel, the project evaluates whether mission-critical alerts can still meet latency and reliability requirements for worker safety and real-time monitoring, extending the underground LoRa performance analysis described in Questions 1–3 into a practical routing and congestion-management study.

6. What specific outcomes are expected by the end of this study?

By the end of this study, the following outcomes are expected:

1. A functioning two-tier LoRa mesh testbed that emulates a deep MRT tunnel environment, with Level 0 sensor nodes and Level 1 LoRaWAN bridge nodes deployed

in a curved, obstructed topology representative of Singapore's deep-tunnel infrastructure.

2. A quantitative performance dataset characterising multi-hop LoRa operation underground, including measurements of RSSI, packet delivery ratio, throughput, and end-to-end latency under both normal and bursty "incident" traffic patterns for gas and temperature telemetry.

3. An empirical evaluation of payload-agnostic forwarding on resource-constrained LoRa mesh relays, quantifying its impact on per-hop forwarding delay, overall channel efficiency, and the ability to meet WSH-driven alert-latency requirements for toxic gas alarms in a structurally obstructed tunnel environment.

4. A set of deployment and configuration guidelines for LoRa mesh in deep-tunnel civil infrastructure (e.g., recommended node spacing, routing metrics, and congestion-management strategies) to support heterogeneous safety-critical sensors without relying on rigid cabling or single-hop LoRaWAN star topologies.

5. A prototype monitoring dashboard (e.g., Raspberry Pi–based) that visualises tunnel topology, node health, and real-time environmental readings, demonstrating how the measured network performance can be operationalised for safety monitoring and incident response in MRT-style tunnel scenarios.

References:

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